PhD Project Outline

The Value of Information in Venture Capital Diligence

Alessandro Guidi · alexguidioff@gmail.com · alessandroguidi.site · linkedin.com/in/alessandroguidi1 · github.com/alexguidioff

---

Target institution

University of St. Gallen, Institute of Technology Management (ITEM-HSG) — Prof. Dr. Petrit Ademi (Entrepreneurial Finance). A methodological co-advisor at ETH (decision-making under uncertainty / optimisation) is envisaged for the modelling side.

1. Research context and focus

Venture investors decide under extreme uncertainty on fragmentary, non-standardised information. Research — including your own conjoint studies of how (corporate) venture capitalists evaluate opportunities — has mapped *which attributes investors weight* when they screen and decide. The question I want to open sits one step earlier and is, so far, largely unasked in venture finance: given a decision, *which missing piece of information — at what cost — most improves it?* Not "which feature predicts success," but "which diligence step is worth doing."

This builds on my accepted WI2026 Student Track paper (*Can Non-Financial Signals Price Private Companies?*), which shows on 3,403 PitchBook deals, under a strict out-of-time holdout, that non-financial signals price private companies competitively with financial baselines — and that **investor-syndicate structure**, not firm financials, is the dominant driver. That last finding connects directly to your work on syndication decisions, and it exposes the ceiling of a purely predictive approach: knowing *what predicts* is not the same as knowing *what to go and learn*.

2. Research questions

Main question: In venture diligence, which information acquisition — at what cost — most improves the quality of the investment decision?

- How can each decision be represented by its *information state*: what a partner actually knows at time *t* (reconstructed point-in-time), before the outcome is realised?
- How do we quantify the *marginal value* of an additional diligence step (a reference call, a cohort metric, a syndicate signal) relative to its cost?
- Which diligence steps do investors appear to under- or over-invest in relative to their information value — and does this differ across deal stages or syndicated vs solo deals?

3. Theoretical framework

The project brings the economics of *value of information* (Howard; Savage) into venture diligence. Where your conjoint experiments identify the *revealed weighting* of attributes in investor decision-making, value-of-information theory asks the normative counterpart: *what would be worth knowing next*. Placing the two side by side — how investors actually weight information vs. how much that information is objectively worth to the decision — is the conceptual core. The technical machinery (Bayesian experimental design / expected information gain; sequential decision-making under uncertainty) exists in the ML literature; its application to venture diligence is open, and it needs a domain anchor in entrepreneurial finance to be posed correctly.

4. Methodology

Feasibility does not depend on proprietary data. Ground truth comes from a *point-in-time reconstruction* of public sources (so the information state at time *t* excludes look-ahead) plus a *virtual-fund simulator* that measures decision quality counterfactually. On top of this I estimate the value of each information type and rank diligence steps by expected value per unit cost, benchmarked against the WI2026 pipeline as a predictive baseline. The design naturally extends to your experimental tradition: a later stage could compare model-implied information value against investors' revealed weighting in a conjoint-style study.

- **Year 1:** formalise the information-state / value-of-information framing; build the point-in-time dataset and the simulator.
- **Year 2:** estimate information value across diligence steps; complete and submit the paper (if it grows: a follow-up on evaluating decision quality by what was knowable, not by the eventual exit).
- **Year 3–4:** extensions (syndication signals, human-vs-model weighting), writing, publications.

5. Expected contribution

The project contributes (i) a *value-of-information framing for venture diligence* — a normative account of which signals are worth acquiring; (ii) a reproducible point-in-time and simulator methodology that makes the question testable without proprietary data; and (iii) empirical evidence, connectable to investors' revealed decision weights, on where diligence effort is mis-allocated. It speaks to entrepreneurial-finance audiences (how investors decide) and generalises beyond VC (PE, credit, M&A), which is what makes it a research line rather than a single study.

6. Fit with host institution

Your research programme — the decision-making of (corporate) venture capitalists, venture screening, and syndication — is exactly the domain this question needs. *Venture Capital Screening* (Drover and Ademi, 2024) frames the very act I want to formalise: what investors examine before committing. Your conjoint experiments on CVC evaluation give a rigorous behavioural counterpart to the normative value-of-information model, and your work on syndication decisions connects to my paper's central finding. I would value ITEM-HSG as the domain home for the project, with an ETH methodological co-advisor for the modelling. I already have an accepted publication in the area and want to build a coherent line of research, not a single paper.

7. Selected references

Howard, R. A. (1966). *Information Value Theory*. IEEE Trans. Systems Science and Cybernetics, 2(1), 22–26.

Savage, L. J. (1954). *The Foundations of Statistics*. Wiley.

Drover, W., & Ademi, P. (2024). *Venture Capital Screening*. Palgrave Macmillan.

Ademi, P., Schuhmacher, M., & Zacharakis, A. (2023). *Evaluating Affordance-Based Opportunities: A Conjoint Experiment of Corporate Venture Capital Managers' Decision-Making*. Entrepreneurship Theory and Practice, 47(6).

Ademi, P., Schuhmacher, M., & Zacharakis, A. (2023). *When Do Sharks Partner Up? An Experiment of Corporate Venture Capital Syndication Decisions.*

Guidi, A., Rashid, S., & Zhong, H. (2026). *Can Non-Financial Signals Price Private Companies?* WI2026 Student Track (accepted).

One-page WI2026 summary attached. Code and manuscript available on request.